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SCHOOL OF COMPUTER SCIENCE & ENGINEERING

WEEKLY PROGRESS REPORT

Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_3CSE413
Project Title	AI-Based smart Traffic congestion prediction and signal optimization.
Reporting period	03 Aug – 14 Aug 2026
Week No.	1

Work Completed This Week: Define the problem of increasing traffic congestion and limitations of fixed time traffic signal. Reviewed existing research gap on traffic congestion prediction. Finalized the project objective scope and proposed approach and select data set from kaggle.

Challenges/Issues:

- Identifying a suitable data set for traffic prediction.
- understanding how ML predictions can support traffic signal optimization.

Plan for Next Week:

- Finalize the system methodology and architecture.
- Begin dataset preprocessing and initial model development

Guide's Remarks:

Satisfactory progress

Signature of Student

Shyamji yadav

Ashoesh kumar

Dev kumar

Bibhuksilpa pradhan

Saurabh kumar rishi

Signature of Guide

Lekha Baghel

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SCHOOL OF COMPUTER SCIENCE & ENGINEERING

WEEKLY PROGRESS REPORT

Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_3CSE413
Project Title	AI-Based Smart Traffic congestion prediction and signal optimization.
Reporting period	17 Aug - 12 sept 2026
Week No.	2

Work Compieted This Week: Designed the workflow of traffic data preprocessing. ML Based speed and traffic-density prediction, and adaptive traffic signal optimization using delhi traffic data set. inspect the data set using. python, pandas.

Challenges/Issues:
- understanding the data set structure and selecting appropriate prediction target.
- Resolved coding erosor during data loading and preprocessing

Plan for Next Week:
- complete initial ML model development.
- Evaluate model performance and save the trained model.

Guide's Remarks:

Satisfactory progress

Signature of Student

shyamji yadav
Ashcesh kumar
Dev Kumar
Bibhukrupa Pradhan
Saurabh Kumar Rishi

Signature of Guide

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SCHOOL OF COMPUTER SCIENCE & ENGINEERING

WEEKLY PROGRESS REPORT

Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_BCSE413
Project Title	AI-Based smart traffic congestion prediction and signal optimization
Reporting period	21 sept - 24 oct 2026
Week No.	3

Work Completed This Week: Implemented and evaluated Regression and classification model of traffic speed and congestion prediction, achieving Random forest MAE of 3.577km/h RMSE of 4.900km/h R^2 of 0.919 and 78% classification accuracy.

Challenges/Issues: observing misclassification between low and medium traffic density categories. Resolving python syntax errors.

Plan for Next Week: complete SUMO installation and verify python-Traci connectivity.
create a four-way intersection with traffic routes and signal phases.

Guide's Remarks: Satisfactory progress

Signature of Student

Shyamji yadav
Ashcesh kumar
Dev kumar
Bibhukosa pradhan
Saurabh kumar Rishi

Signature of Guide